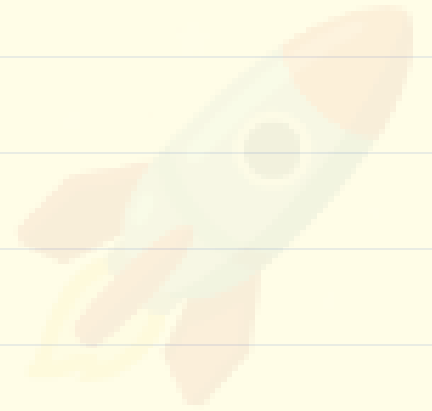


Your DevOps Cheat Sheet

Zero → Ship-Ready



-  1.2k stars · 7.4k forks · community-built since 2023
-  One task/day. One real outcome/day. No theory dumps.
-  Linux → Git → Docker → K8s → CI/CD → Terraform → Ansible → AWS → DevSecOps
-  Slide 7:  the mistake everyone makes about DevOps

"Not a course. A discipline + execution challenge." — repo README

Linux & Git — everything lives here



- ✎ Know your permissions: `chmod 755`, `chown`, `umask`
- ✎ Shell scripts = your first automation. If you do it twice, script it.
- ✎ `ps aux | grep`, `kill -9`, `top` — process management basics
- ✎ Git: branch per feature, commit messages that explain *why*, not what
- ✎ `git rebase` rewrites history. `git merge` preserves it. Know when you want which.

★ `/var/log/syslog` and `journalctl -xe` are your first stops when prod breaks at 2am.

Docker & Ansible — pack it, repeat it

- ✎ **Docker:** image is immutable. Container is throwaway. That's the whole model.
- ✎ Multi-stage builds: dev → build → final slim image. Don't ship build tools to prod.
- ✎ Combine `RUN apt-get update && install` in ONE layer or cache breaks on you.
- ✎ **Ansible:** idempotent = run it 10 times, same result. That's not optional, that's the point.
- ✎ Use Roles, not giant playbooks. Structure: `tasks/`, `handlers/`, `vars/`

★ *SSH key auth between control + worker nodes trips everyone up the first time. Set it up properly before running any playbook.*

Kubernetes — containers at scale



- ✎ Pod = smallest unit. Deployment = manages Pods. Service = exposes them. Ingress = routes in.
- ✎ ConfigMap for config. Secret for credentials. base64 ≠ encryption — don't get lazy here.
- ✎ HPA autoscales on CPU/memory. Set `requests` + `limits` or HPA can't do its job.
- ✎ Rolling update = zero downtime by default. `maxUnavailable: 0` + `maxSurge: 1` for safety.
- ✎ PersistentVolume + PVC = storage that survives pod restarts. Databases need this.

★ *CrashLoopBackOff* = "your app keeps dying." Check: `kubectl logs <pod> --previous`

CI/CD & Terraform — automate everything



- ✎ **GitHub Actions:** every push triggers the pipeline. Stages: lint → test → build → scan → deploy.
- ✎ Secrets in `${{ secrets.NAME }}`. Never hardcode. Gitleaks will find it anyway.
- ✎ **Terraform:** `for_each` on `map(object())` > `count` when resources need distinct roles.
- ✎ Always run `terraform plan` before `apply`. Remote state in S3 + DynamoDB lock — non-negotiable in teams.
- ✎ Argo CD = GitOps. Git is the source of truth. Cluster state follows Git — always.

★ *A pipeline with no test stage is just a fancy way to push bugs faster.*

AWS & DevSecOps — build it secure

-  **AWS core:** EC2, S3, IAM, VPC, RDS — understand these before anything else.
-  IAM: least privilege always. If you gave `*` permissions "just to test" — revoke it now.
-  **Trivy** = scan container images. **Gitleaks** = secrets in code. **Bandit** = Python vulns.
-  **Hadolint** for Dockerfiles. Bad Dockerfile habits = prod vulnerabilities.
-  Shift left = scan in the pipeline, before anything hits production. Not after.

★ *pip-audit catches vulnerable Python deps. Add it to your pipeline — 5 mins to set up, could save hours of incident response.*

⚠ THE MISTAKE EVERYONE MAKES — READ THIS TWICE

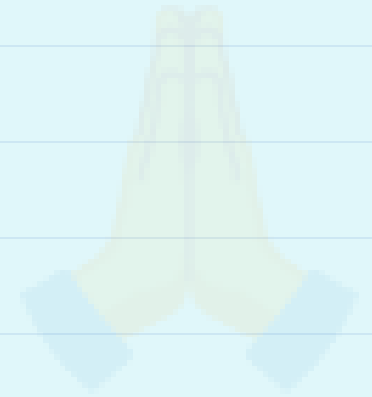
DevOps is NOT a tool. Stop. Treating. It. Like. One.

- ✗ *"We installed Jenkins — we're doing DevOps now."* No. That's a CI tool.
- ✗ *"We hired a DevOps engineer."* One person can't fix a culture problem.
- ✗ *"DevOps is just Docker + K8s."* Tools serve the culture. Not the other way.
- ✓ DevOps = Dev and Ops sharing ownership. Faster feedback. Smaller batches. No blame.
- ✓ The 90DaysOfDevOps repo itself is DevOps in action — open, collaborative, community-owned.

★ *If devs and ops still blame each other when prod breaks — no pipeline will fix that. Fix the culture first. The tools follow.*

90 days. 90 lessons.

One repo. One community. ❤️



- 🌱 Started asking "what's a container?" — left running K8s microservices on kind in WSL.
- 🌱 Built real pipelines. Broke them. Fixed them. Documented every broken step publicly.
- 🌱 The repo: **1.2k** ★ · **7.4k forks** · **642 commits** — built by hundreds of people like you.
- 🌱 This is not the finish line. It's the baseline. Keep building in public.

★ github.com/TrainWithShubham/90DaysOfDevOps

*Save this. Share it. Pass it on. — Prakhar Srivastava | #90DaysOfDevOps |
#TrainWithShubham*